

CS3481 - Tutorial & Exercise

Tutorial 1 (Data)

(1) Simple Matching Coefficient (SMC):

$$SMC = \frac{f_{11} + f_{00}}{f_{11} + f_{10} + f_{01} + f_{00}}$$

Jaccard Coefficient:

$$J = \frac{f_{11}}{f_{11} + f_{10} + f_{01}}$$

1. Calculate the value of the Simple Matching Coefficient and the Jaccard coefficient for the two vectors $x=(1,0,0,0,0,1,1,0)$ and $y=(0,0,1,0,1,0,1,0)$.
2. What is the main difference between these two measures?

Answers:

$$f_{11} = 1, f_{00} = 3, f_{10} = 2, f_{01} = 2$$

$$SMC = \frac{1+3}{1+2+2+3} = \frac{4}{8} = 0.5$$

$$J = \frac{1}{1+2+2} = \frac{1}{5} = 0.2$$

The main difference is that Jaccard coefficient does not consider the cases where both attributes are 0 (i.e., f_{00}) in its calculation, while Simple Matching Coefficient does. This makes Jaccard more suitable for situations where the presence of a feature is more important than its absence.

(2)

Article	economics	market	bank	company	medicine	insurance
1	3	5	2	-	-	-
2	12	15	10	8	-	-
3	-	-	-	5	3	5

$$\text{Cosine similarity between articles 1 and 2: } S_c(A_1, A_2) = \frac{A_1 \cdot A_2}{\|A_1\| \cdot \|A_2\|} = 0.920483$$

$$\text{Cosine similarity between articles 2 and 3: } S_c(A_2, A_3) = \frac{A_2 \cdot A_3}{\|A_2\| \cdot \|A_3\|} = 0.225564$$

Observation: Articles 1 and 2 are more similar to each other than articles 2 and 3, as indicated by the higher cosine similarity value.

(3)

Article	Words
1	dollar:1, industry:4, country:2, loan:3, deal:2, government:2, very:6
2	machinery:2, labor: 3, market:4, industry:2, government:3, very:5
3	job:5, inflation:3, company:2, market:3, country:2, index:3, very:8
4	domestic:3, forecast:2, government:1, market:2, sale:3, price:2, very:2
5	patient:4, symptom:2, drug:3, health:2, clinic:2, doctor:2, very:5
6	pharmaceutical:2, company:3, drug:2, vaccine:1, flu:3, very:3

Article	Words
7	death:2, cancer:4, drug:3, government:4, health:3, director:2, very:7
8	medical:2, cost:3, government:2, patient:2, health:3, care:1, very:5

TF-IDF formula:

$$t_{ij} = \text{tf}_{ij} \times \log_2 \left(\frac{N}{\text{df}_j} \right)$$

Where: tf_{ij} = term frequency of term j in document i, N = total number of documents, df_j = document frequency of term j.

Apply this transformation to (i) the count of the word "pharmaceutical" in article 6, (ii) the count of the word "government" in article 2, and (iii) the count of the word "very" in article 5. What is your observation? What is the purpose of this transformation?

Answers:

$$(i) t_{6,\text{pharmaceutical}} = 2 \times \log_2 \left(\frac{8}{1} \right) = 6$$

$$(ii) t_{2,\text{government}} = 3 \times \log_2 \left(\frac{8}{5} \right) = 2.034$$

$$(iii) t_{5,\text{very}} = 5 \times \log_2 \left(\frac{8}{8} \right) = 0$$

Observation: The term "pharmaceutical" has a high TF-IDF score in article 6, indicating its significance in that document. The term "government" has a moderate score in article 2, reflecting its relevance but also its presence in other documents. The term "very" has a TF-IDF score of 0 in article 5, suggesting it is a common word across all documents and does not contribute to distinguishing the content of the articles.

Purpose: The TF-IDF transformation helps to highlight important words in a document while down-weighting common words that appear frequently across all documents. This allows for better identification of unique and relevant terms that contribute to the meaning of each document.

Tutorial 2 (Data)

(1) Derive the mathematical relationship between cosine similarity and Euclidean distance when each data object has an L2 length of 1.

Answers:

$$\begin{aligned}
 D(x, y) &= \sqrt{\sum_{i=1}^n (x_i - y_i)^2} \\
 &= \sqrt{\sum_{i=1}^n (x_i^2 - 2x_i y_i + y_i^2)} \\
 &= \sqrt{\sum_{i=1}^n x_i^2 + \sum_{i=1}^n y_i^2 - 2 \sum_{i=1}^n x_i y_i} \\
 &= \sqrt{\|x\|^2 + \|y\|^2 - 2(x \cdot y)} \\
 &= \sqrt{1 + 1 - 2(x \cdot y)} \\
 &= \sqrt{2(1 - S_c(x, y))}
 \end{aligned}$$

(2) We consider the following similarity measure between two vectors x and y.

$$\sigma(x, y) = \frac{x \cdot y}{(|x|^2 + |y|^2 - x \cdot y)}$$

Show that this measure corresponds to the Jaccard coefficient when x and y are binary vectors.

Answers:

For binary vectors, the dot product $x \cdot y$ counts the number of dimensions where both vectors have a value of 1 (denoted as f_{11}). The term $|x|^2$ counts the number of dimensions where vector x has a value of 1 (denoted as $f_{10} + f_{11}$), and similarly for $|y|^2$ (denoted as $f_{01} + f_{11}$).

$$\text{So } f_{11} + f_{10} + f_{01} = (f_{10} + f_{11}) + (f_{01} + f_{11}) - f_{11} = |x|^2 + |y|^2 - x \cdot y.$$

$$\text{So } J(x, y) = \frac{f_{11}}{f_{11} + f_{10} + f_{01}} = \frac{x \cdot y}{(|x|^2 + |y|^2 - x \cdot y)} = \sigma(x, y)$$

(3) We consider the following data points: (2, 19), (9, 6), (7, 15), (5, 12).

1. Calculate the covariance matrix of this set of data.
2. Calculate the correlation coefficient between the two attributes.

Answers:

$$\bar{x} = \frac{2+9+7+5}{4} = 5.75, \bar{y} = \frac{19+6+15+12}{4} = 13$$

$$\text{Var}(x) = \frac{1}{4-1}((2-5.75)^2 + (9-5.75)^2 + (7-5.75)^2 + (5-5.75)^2) = 8.9167$$

$$\text{Var}(y) = \frac{1}{4-1}((19-13)^2 + (6-13)^2 + (15-13)^2 + (12-13)^2) = 30$$

$$\text{Cov}(X, Y) = \frac{1}{4-1}((2-5.75)(19-13) + (9-5.75)(6-13) + (7-5.75)(15-13) + (5-5.75)(12-13)) = -14$$

$$C = \begin{bmatrix} 8.9167 & -14 \\ -14 & 30 \end{bmatrix}$$

$$\text{Correlation coefficient: } r_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{-14}{\sqrt{8.9167 \times 30}} = -0.856$$

Exercise 1 (Data & Decision Tree)

(1) Discuss whether or not each of the following activities is a data mining task.

1. Computing the total sales of a company.
2. Sorting a student database based on student identification numbers.
3. Predicting the future stock price of a company using historical records.

Answers:

1. No, a straightforward aggregation task is at data exploration level.
2. No, sorting is a basic data processing task.
3. Yes, this is a predictive modeling task.

(2) We consider a collection of news articles shown in the following table. Each article is represented as a set of word-frequency pairs (w,c), where w is a word and c is the number of times the word appears in the article.

Suppose we apply cluster analysis to group the set of articles based on their respective topics. Suggest a suitable representation format for each article such that the degree of similarity between two articles can be readily compared.

Answers:

A suitable representation format for each article is the vector space model, where each article is represented as a vector in a high-dimensional space. Each dimension corresponds to a unique word from the entire collection of articles, and the value in each dimension represents the frequency of that word in the article. This allows for easy computation of similarity measures, such as L2-norm or cosine similarity, between articles based on their vector representations.

(3) We consider the problem of predicting whether a loan applicant will default on his/her loan. A data set for this problem is shown in the table in the next page. Each record contains the personal information of a borrower, along with a class label indicating whether the borrower has defaulted on loan payments.

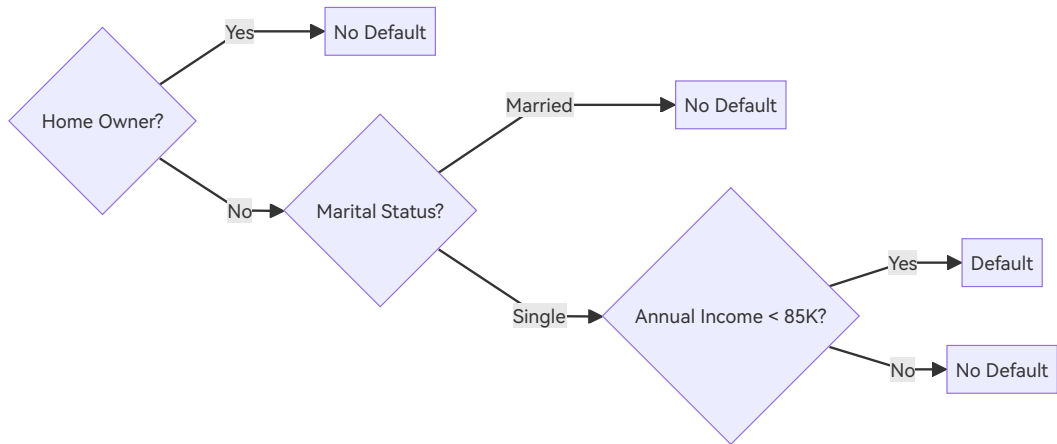
Applicant No.	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	Yes
4	Yes	Married	120K	No
5	No	Single	95K	No
6	No	Married	60K	No
7	Yes	Single	220K	No
8	No	Single	80K	Yes
9	No	Married	75K	No
10	No	Single	90K	No

1. Suppose we first focus on the attribute Home Owner. Describe in what way this attribute can help us to determine whether a borrower will default on his/her loan payment or not.
2. Suppose we next focus on the subset of non-home owners. Describe in what way the attribute Marital Status can help us to determine whether a borrower will default or not.
3. Finally, determine how we can make use of the attribute Annual Income to complete the prediction process.

Answers:

1. Let $A = \text{Home Owner} = \text{Yes}$ and $B = \text{Defaulted Borrower} = \text{Yes}$. It can be observed that all home owners (Applicants 1, 4, and 7) did not default on their loans. Therefore, home owners can successfully repay their loans, while more information is needed for non-home owners.
2. Let $C = \text{Marital Status} = \text{Single}$ and we only consider non-home owners. Among these, Applicants 3 and 8 (single) defaulted, Applicants 5 and 10 (single) did not default, and Applicants 2, 6, and 9 (married) did not default. Therefore, married non-home owners can successfully repay their loans, while more information is needed for single non-home owners.
3. Let $E = \text{Annual Income} < 85\text{K}$. Among Applicants 3, 5, 8, 10, we can observe that Applicants 3 and 8 (with income less than 85K) defaulted, while Applicants 5 and 10 (with income greater than or equal to 85K) did not default. Therefore, if his/her annual income is greater than or equal to 85K, they can successfully repay their loans; if his/her annual income is less than 85K, they cannot successfully repay their loans.

Decision Tree:



Tutorial 3 (Decision Tree)

(1) Explain why there are $2^{S-1} - 1$ ways of creating a binary partition for a nominal attribute with S distinct values.

Answers:

For a nominal attribute with S distinct values, we can think of each value as a unique category. To create a binary partition, we need to decide which categories will go into one group and which will go into the other group.

This gives us 2^S possible ways to assign the S categories into two groups (since each category can either be in group 1 or group 2). However, we need to exclude the cases where all categories are in one group and the other group is empty, which accounts for 2 cases (all in group 1 or all in group 2). Additionally, group 1 and group 2 are interchangeable (i.e., swapping the groups does not create a new partition), so we need to divide by 2 to account for this symmetry.

Thus, the number of unique binary partitions is given by:

$$\frac{2^S - 2}{2} = 2^{S-1} - 1$$

(2) We consider the training examples shown in the following table for a binary classification problem.

Instance	a_1	a_2	a_3	Class
1	T	T	1	+
2	T	T	6	+
3	T	F	5	-
4	F	F	4	+
5	F	T	7	-
6	F	T	3	-
7	F	F	8	-
8	T	F	7	+
9	F	T	5	-

(a) What is the original entropy of this set of training instances?

(b) What are the information gains when a_1 and a_2 are used for partitioning the training set respectively?

Answers:

(a) The original entropy is :

$$E = - \left(\frac{5}{9} \log_2 \frac{5}{9} + \frac{4}{9} \log_2 \frac{4}{9} \right) = 0.991$$

(b) The new entropy after partitioning by a_1 is (T: 3+ 1-, F: 1+ 4-):

$$E_{a_1} = \frac{4}{9} \left(-\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} \right) + \frac{5}{9} \left(-\frac{1}{5} \log_2 \frac{1}{5} - \frac{4}{5} \log_2 \frac{4}{5} \right) = 0.762$$

The information gain for a_1 is: $IG_{a_1} = E - E_{a_1} = 0.991 - 0.762 = 0.229$

The new entropy after partitioning by a_2 is (T: 2+ 3-, F: 2+ 2-):

$$E_{a_2} = \frac{5}{9} \left(-\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} \right) + \frac{4}{9} \left(-\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} \right) = 0.984$$

The information gain for a_2 is: $IG_{a_2} = E - E_{a_2} = 0.991 - 0.984 = 0.007$

(3)

Condition	TEMP	HUMID	AIR	Number
1	High	High	Low	Large
2	High	High	High	Large
3	Medium	High	Low	Large
4	Low	Low	Low	Small
5	Low	Low	High	Large
6	Medium	Low	Low	Small
7	Medium	Low	High	Large
8	Medium	High	High	Large
9	High	Low	Low	Large
10	High	Low	High	Large

Entropy of the original set: $E = - \left(\frac{8}{10} \log_2 \frac{8}{10} + \frac{2}{10} \log_2 \frac{2}{10} \right) = 0.722$

For the root split:

TEMP (High: 4+ 0-, Medium: 3+ 1-, Low: 1+ 1-):

$$IG_T = E - \left[\frac{4}{10}(0) + \frac{4}{10} \left(-\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} \right) + \frac{2}{10} \left(-\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} \right) \right] = 0.722 - 0.525 = 0.19$$

HUMID (High: 4+ 0-, Low: 4+ 2-):

$$IG_H = E - \left[\frac{4}{10}(0) + \frac{6}{10} \left(-\frac{4}{6} \log_2 \frac{4}{6} - \frac{2}{6} \log_2 \frac{2}{6} \right) \right] = 0.722 - 0.551 = 0.171$$

AIR (High: 5+ 0-, Low: 3+ 2-):

$$IG_A = E - \left[\frac{5}{10}(0) + \frac{5}{10} \left(-\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} \right) \right] = 0.722 - 0.485 = 0.237$$

Since IG_A is the highest, we choose AIR as the root node.

$$E_{AIR} = - \left(\frac{3}{5} \log_2 \frac{3}{5} + \frac{2}{5} \log_2 \frac{2}{5} \right) = 0.971$$

(Note: The definition of "entire set" changes after a split, so the entropy need to be recalculated.)

The high branch is pure. For AIR = Low:

Condition	TEMP	HUMID	AIR	Number
1	High	High	Low	Large
3	Medium	High	Low	Large
4	Low	Low	Low	Small
6	Medium	Low	Low	Small
9	High	Low	Low	Large

TEMP (High: 2+ 0-, Medium: 1+ 1-, Low: 0+ 1-):

$$IG_T = E_{AIR} - \left[\frac{2}{5}(0) + \frac{2}{5}\left(-\frac{1}{2}\log_2 \frac{1}{2} - \frac{1}{2}\log_2 \frac{1}{2}\right) + \frac{1}{5}(0) \right] = 0.971 - 0.4 = 0.571$$

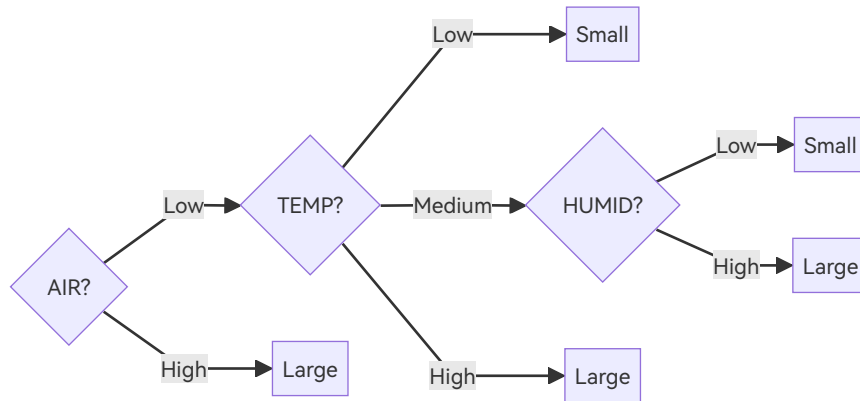
HUMID (High: 2+ 0-, Low: 1+ 2-):

$$IG_H = E_{AIR} - \left[\frac{2}{5}(0) + \frac{3}{5}\left(-\frac{1}{3}\log_2 \frac{1}{3} - \frac{2}{3}\log_2 \frac{2}{3}\right) \right] = 0.971 - 0.551 = 0.42$$

Since IG_T is the highest, we choose TEMP as the second node.

High and Small branches are pure. For TEMP = Medium, split by HUMID: High = Large, Low = Small.

The decision tree is shown below:



Tutorial 4 (Decision Tree)

Continue from Tutorial 3 (2)

(a) Calculate the respective changes in the Gini index value when a_1 and a_2 are used for partitioning the training set.

The original Gini index is :

$$G = 1 - \left(\frac{5}{9}\right)^2 - \left(\frac{4}{9}\right)^2 = 0.494$$

The new Gini index after partitioning by a_1 is (T: 3+ 1-, F: 1+ 4-):

$$G_{a_1} = \frac{4}{9}\left(1 - \left(\frac{3}{4}\right)^2 - \left(\frac{1}{4}\right)^2\right) + \frac{5}{9}\left(1 - \left(\frac{1}{5}\right)^2 - \left(\frac{4}{5}\right)^2\right) = 0.344$$

The change in Gini index for a_1 is: $\Delta G_{a_1} = G - G_{a_1} = 0.494 - 0.344 = 0.150$

The new Gini index after partitioning by a_2 is (T: 2+ 3-, F: 2+ 2-):

$$G_{a_2} = \frac{5}{9}\left(1 - \left(\frac{2}{5}\right)^2 - \left(\frac{3}{5}\right)^2\right) + \frac{4}{9}\left(1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2\right) = 0.489$$

The change in Gini index for a_2 is: $\Delta G_{a_2} = G - G_{a_2} = 0.494 - 0.489 = 0.005$

(b) Calculate the respective changes in the classification error rate when a_1 and a_2 are used for partitioning the training set.

The original classification error rate is :

$$E = 1 - \max\left(\frac{5}{9}, \frac{4}{9}\right) = 0.444$$

The new classification error rate after partitioning by a_1 is (T: 3+ 1-, F: 1+ 4-):

$$E_{a_1} = \frac{4}{9}\left(1 - \max\left(\frac{3}{4}, \frac{1}{4}\right)\right) + \frac{5}{9}\left(1 - \max\left(\frac{1}{5}, \frac{4}{5}\right)\right) = 0.222$$

The change in classification error rate for a_1 is: $\Delta E_{a_1} = E - E_{a_1} = 0.444 - 0.222 = 0.222$

The new classification error rate after partitioning by a_2 is (T: 2+ 3-, F: 2+ 2-):

$$E_{a_2} = \frac{5}{9}\left(1 - \max\left(\frac{2}{5}, \frac{3}{5}\right)\right) + \frac{4}{9}\left(1 - \max\left(\frac{2}{4}, \frac{2}{4}\right)\right) = 0.444$$

The change in classification error rate for a_2 is: $\Delta E_{a_2} = E - E_{a_2} = 0.444 - 0.444 = 0$

(c) For a_3 , compute the information gain for every possible split. What is the best threshold for splitting the set of attribute values?

Splitting at 1.5: (<: 1+ 0-, >: 3+ 5-):

$$E_{1.5} = \frac{1}{9}\left(-\frac{1}{1}\log_2 \frac{1}{1}\right) + \frac{8}{9}\left(-\frac{3}{8}\log_2 \frac{3}{8} - \frac{5}{8}\log_2 \frac{5}{8}\right) = 0.848, IG_{1.5} = 0.143$$

Splitting at 3.5: (<: 1+ 1-, >: 3+ 4-):

$$E_{3.5} = \frac{2}{9}\left(-\frac{1}{2}\log_2 \frac{1}{2} - \frac{1}{2}\log_2 \frac{1}{2}\right) + \frac{7}{9}\left(-\frac{3}{7}\log_2 \frac{3}{7} - \frac{4}{7}\log_2 \frac{4}{7}\right) = 0.989, IG_{3.5} = 0.002$$

Splitting at 4.5: (<: 2+ 1-, >: 2+ 4-):

$$E_{4.5} = \frac{3}{9}\left(-\frac{2}{3}\log_2 \frac{2}{3} - \frac{1}{3}\log_2 \frac{1}{3}\right) + \frac{6}{9}\left(-\frac{2}{6}\log_2 \frac{2}{6} - \frac{4}{6}\log_2 \frac{4}{6}\right) = 0.918, IG_{4.5} = 0.073$$

Splitting at 5.5: (<: 2+ 3-, >: 2+ 2-):

$$E_{5.5} = \frac{5}{9}\left(-\frac{2}{5}\log_2 \frac{2}{5} - \frac{3}{5}\log_2 \frac{3}{5}\right) + \frac{4}{9}\left(-\frac{2}{4}\log_2 \frac{2}{4} - \frac{2}{4}\log_2 \frac{2}{4}\right) = 0.984, IG_{5.5} = 0.007$$

Splitting at 6.5: (<: 3+ 3-, >: 1+ 2-):

$$E_{6.5} = \frac{6}{9}\left(-\frac{3}{6}\log_2 \frac{3}{6} - \frac{3}{6}\log_2 \frac{3}{6}\right) + \frac{3}{9}\left(-\frac{1}{3}\log_2 \frac{1}{3} - \frac{2}{3}\log_2 \frac{2}{3}\right) = 0.973, IG_{6.5} = 0.018$$

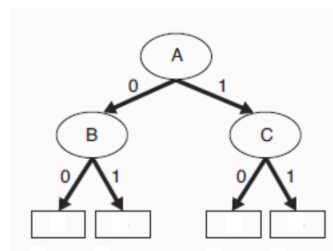
Splitting at 7.5: (<: 4+ 4-, >: 0+ 1-):

$$E_{7.5} = \frac{8}{9}\left(-\frac{4}{8}\log_2 \frac{4}{8} - \frac{4}{8}\log_2 \frac{4}{8}\right) + \frac{1}{9}\left(-\frac{1}{1}\log_2 \frac{1}{1}\right) = 0.889, IG_{7.5} = 0.102$$

So the best threshold for splitting the set of attribute values is 1.5.

Tutorial 5 (Classifier Evaluation)

(1)



Training:

Instance	A	B	C	Class
1	0	0	0	+
2	0	0	1	+
3	0	1	0	+
4	0	1	1	-
5	1	0	0	+
6	1	0	0	+
7	1	1	0	-
8	1	0	1	+
9	1	1	0	-
10	1	1	0	-

Instance	A	B	C	Class
1	0	0	0	+
2	0	0	1	+
3	0	1	0	+
4	0	1	1	-
5	1	0	0	+

Instance	A	B	C	Class
6	1	0	0	+
7	1	1	0	-
8	1	0	1	+
9	1	1	0	-
10	1	1	0	-

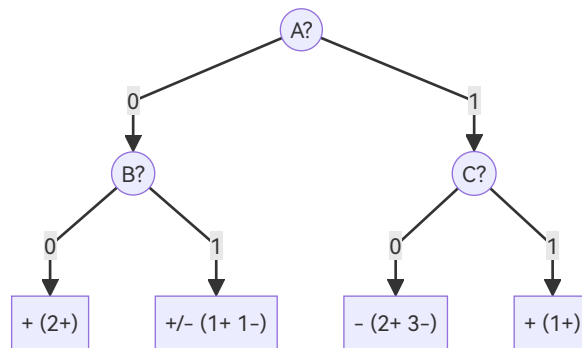
(a) Estimate the generalization error rate of the tree by using the resubstitution estimate.

(b) Estimate the generalization error rate by using a penalty term of 0.5 for each leaf node.

Answers:

- Node A: 0 (3+ 1-), 1 (3+ 3-)
- Node B (A=0): 0 (2+), 1 (1+ 1-)
- Node C (A=1): 0 (2+ 3-), 1 (1+)

So the decision tree can be represented as follows:



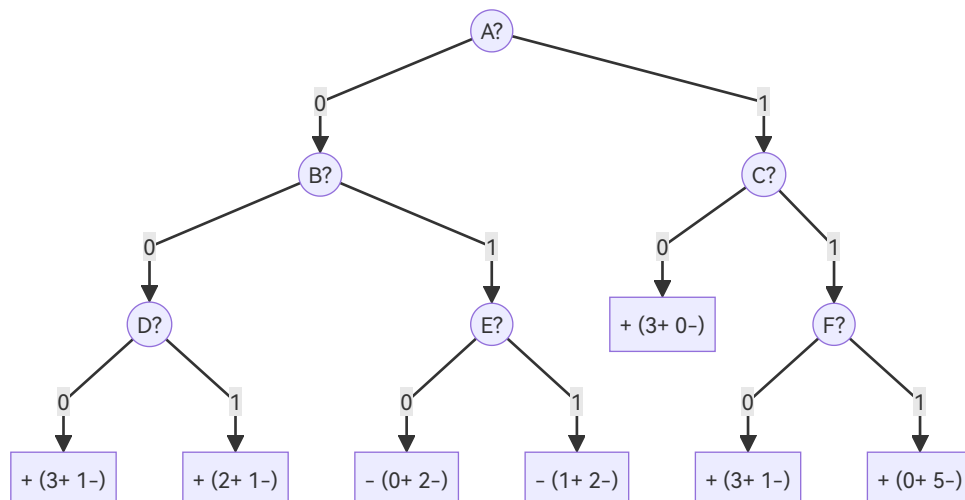
(a) The resubstitution estimate means that we use the training error rate to estimate the generalization error rate. The training error rate is :

$$E_{resub} = \frac{3}{10} = 0.3$$

(b) The penalty term for each leaf node is 0.5, and there are 4 leaf nodes in the tree. So the total penalty is $0.5 \times 4 = 2$. The adjusted error rate is :

$$E_{adjusted} = E_{resub} + \frac{2}{10} = 0.3 + 0.2 = 0.5$$

(2)



Suppose a penalty term of 1.5 is assigned to each leaf node.

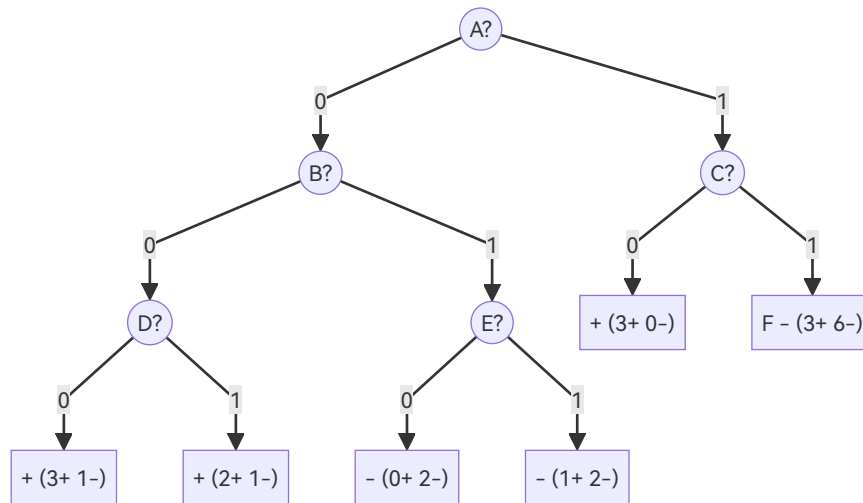
(a) Estimate the generalization error rate if the sub-tree associated with node F is pruned and replaced with a leaf node.

(b) Estimate the generalization error rate if the sub-trees associated with nodes D and E are pruned and replaced with leaf nodes.

(c) Estimate the generalization error rate if the above operations are performed together.

Answers:

(a) If the sub-tree associated with node F is pruned, the new tree will be:



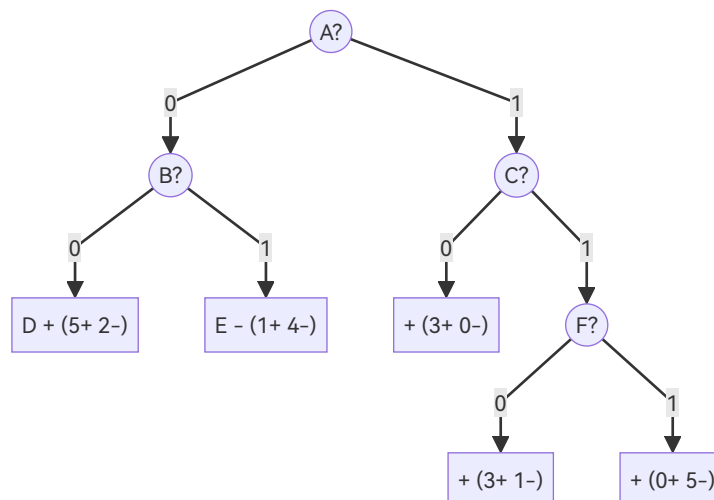
Total cases in the training set: 24

Number of errors in the new tree: 1 ("3+ 1-") + 1 ("2+ 1-") + 1 ("1+ 2-") + 3 ("3+ 6-") = 6

Number of leaves in the new tree: 6

Error rate after pruning F: $E_{pruneF} = \frac{6}{24} + \frac{1.5 \times 6}{24} = 0.625$

(b) If the sub-trees associated with nodes D and E are pruned, the new tree will be:

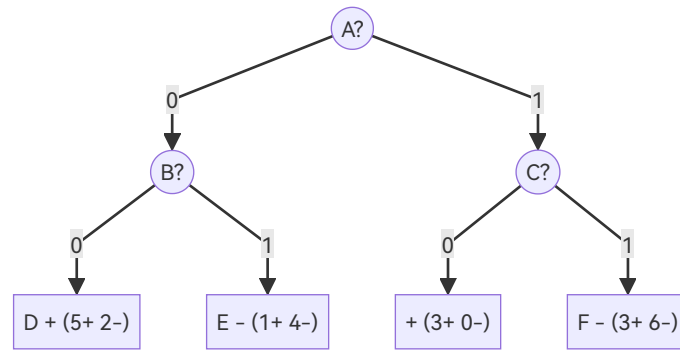


Number of errors in the new tree: 2 ("5+ 2-") + 1 ("1+ 4-") + 1 ("3+ 1-") = 4

Number of leaves in the new tree: 5

Error rate after pruning D and E: $E_{pruneDE} = \frac{4}{24} + \frac{1.5 \times 5}{24} = 0.479$

(c) If both operations are performed together, the new tree will be:



Number of errors in the new tree: 2 ("5+ 2-") + 1 ("1+ 4-") + 3 ("3+ 6-") = 6

Number of leaves in the new tree: 4

Error rate after pruning D, E, and F: $E_{pruneDEF} = \frac{6}{24} + \frac{1.5 \times 4}{24} = 0.5$

Tutorial 6 (Probabilistic Classifier)

(1) For the loan default prediction problem in the lecture notes, verify the class conditional probability values of the categorical attributes Home Owner and Marital Status, and also the sample mean and variance values of the attribute Annual Income for the class Yes.

Tid	Home Owner	Marital Status	Annual Income	Defaulted Borrower
1	Yes	Single	125K	No
2	No	Married	100K	No
3	No	Single	70K	No
4	Yes	Married	120K	No
5	No	Divorced	95K	Yes
6	No	Married	60K	No
7	Yes	Divorced	220K	No
8	No	Single	85K	Yes
9	No	Married	75K	No
10	No	Single	90K	Yes

Answer:

$$P(\text{Home Owner} = \text{Yes} | \text{Yes}) = \frac{0}{3} = 0$$

$$P(\text{Home Owner} = \text{No} | \text{Yes}) = \frac{3}{3} = 1$$

$$P(\text{Home Owner} = \text{Yes} | \text{No}) = \frac{3}{7}$$

$$P(\text{Home Owner} = \text{No} | \text{No}) = \frac{4}{7}$$

$$P(\text{Marital Status} = \text{Single} | \text{Yes}) = \frac{2}{3}$$

$$P(\text{Marital Status} = \text{Married} | \text{Yes}) = \frac{0}{3} = 0$$

$$P(\text{Marital Status} = \text{Divorced} | \text{Yes}) = \frac{1}{3}$$

$$P(\text{Marital Status} = \text{Single} | \text{No}) = \frac{2}{7}$$

$$P(\text{Marital Status} = \text{Married} | \text{No}) = \frac{4}{7}$$

$$P(\text{Marital Status} = \text{Divorced} | \text{No}) = \frac{1}{7}$$

$$\bar{x}_{\text{Yes}} = \frac{95+85+90}{3} = 90$$

$$s_{\text{Yes}}^2 = \frac{(95-90)^2 + (85-90)^2 + (90-90)^2}{3-1} = 25$$

$$s_{\text{Yes}} = \sqrt{25} = 5$$

Note: sample variance is used.

(2) Continue from (1). Predict the class label of a test record: Home Owner = No, Marital Status = Married, Annual Income = 120K.

$$\bar{x}_{\text{No}} = \frac{125+100+70+120+60+220+75}{7} = 110$$

$$s_{\text{No}}^2 = \frac{(125-110)^2 + (100-110)^2 + (70-110)^2 + (120-110)^2 + (60-110)^2 + (220-110)^2 + (75-110)^2}{7-1} = 2975$$

$$s_{\text{No}} = \sqrt{2975} \approx 54.54$$

$$P(\text{Annual Income} = X) = \frac{1}{\sqrt{2\pi}s} \exp\left(-\frac{(X-\bar{x})^2}{2s^2}\right)$$

$$P(\text{Annual Income} = 120K|\text{Yes}) = \frac{1}{\sqrt{2\pi} \times 5} \exp\left(-\frac{(120-90)^2}{2 \times 25}\right) \approx 1.22 \times 10^{-9}$$

$$P(\text{Annual Income} = 120K|\text{No}) = \frac{1}{\sqrt{2\pi} \times 54.54} \exp\left(-\frac{(120-110)^2}{2 \times 2975}\right) \approx 0.007192$$

$$\text{Prior probabilities: } P(\text{Yes}) = \frac{3}{10} = 0.3, P(\text{No}) = \frac{7}{10} = 0.7$$

Posterior probabilities:

$$P(\text{Yes}|X) = P(\text{Home Owner} = \text{No}|\text{Yes}) \times P(\text{Marital Status} = \text{Married}|\text{Yes}) \times P(\text{Annual Income} = 120K|\text{Yes})$$

$$P(\text{No}|X) = P(\text{Home Owner} = \text{No}|\text{No}) \times P(\text{Marital Status} = \text{Married}|\text{No}) \times P(\text{Annual Income} = 120K|\text{No})$$

Since $P(\text{No}|X) > P(\text{Yes}|X)$, the predicted class label for the test record is No.

(3)

Record	A	B	C	Class
1	F	F	F	+
2	F	F	T	-
3	F	T	T	-
4	F	T	T	-
5	F	F	F	+
6	T	F	F	+
7	T	F	T	-
8	T	F	T	-
9	T	T	T	+
10	T	F	T	+

(a) Estimate the class-conditional probabilities.

(b) Predict the class label for a test example (A=F, B=T, C=F) using the naïve Bayes approach.

Answers:

(a)

$$P(A = F|+) = \frac{2}{5}, P(A = T|+) = \frac{3}{5}$$

$$P(A = F|-) = \frac{3}{5}, P(A = T|-) = \frac{2}{5}$$

$$P(B = F|+) = \frac{4}{5}, P(B = T|+) = \frac{1}{5}$$

$$P(B = F|-) = \frac{3}{5}, P(B = T|-) = \frac{2}{5}$$

$$P(C = F|+) = \frac{3}{5}, P(C = T|+) = \frac{2}{5}$$

$$P(C = F|-) = 0, P(C = T|-) = 1$$

(b)

$$P(+|A = F, B = T, C = F) = \frac{P(A=F|+) \times P(B=T|+) \times P(C=F|+) \times P(+)}{P(A=F) \times P(B=T) \times P(C=F)} = \frac{\frac{2}{5} \times \frac{1}{5} \times \frac{3}{5} \times 0.5}{P(A=F) \times P(B=T) \times P(C=F)} = \frac{0.024}{p}$$

$$P(-|A = F, B = T, C = F) = \frac{P(A=F|-) \times P(B=T|-) \times P(C=F|-) \times P(-)}{P(A=F) \times P(B=T) \times P(C=F)} = \frac{\frac{3}{5} \times \frac{2}{5} \times 0 \times 0.5}{P(A=F) \times P(B=T) \times P(C=F)} = 0$$

Since $P(+|A = F, B = T, C = F) > P(-|A = F, B = T, C = F)$, the predicted class label for the test example is +.

Tutorial 7 (K-means Clustering)

(1) Consider the following set of one-dimensional points: {6, 12, 18, 24, 30, 42, 48}.

(a) For each of the following sets of initial centroids (i) {18,45}, (ii) {15,40}, create two clusters by assigning each point to the nearest centroid, and then calculate the sum squared error for each set of two clusters after updating the centroids.

(b) Do both sets of centroids represent stable solutions, i.e., if the K-means algorithm is applied to this set of points using the given centroids as the starting centroids, would there be any change in the clusters generated?

Answers:

(i) Initial centroids: {18, 45}

- Cluster 1 (centroid = 18): {6, 12, 18, 24, 30}
- Cluster 2 (centroid = 45): {42, 48}

$$\text{Updated centroids: Cluster 1: } \frac{6+12+18+24+30}{5} = 18, \text{ Cluster 2: } \frac{42+48}{2} = 45$$

Sum squared error: Cluster 1 = 360, Cluster 2 = 18, Total = 378

(ii) Initial centroids: {15, 40}

- Cluster 1 (centroid = 15): {6, 12, 18, 24}
- Cluster 2 (centroid = 40): {30, 42, 48}

$$\text{Updated centroids: Cluster 1: } \frac{6+12+18+24}{4} = 15, \text{ Cluster 2: } \frac{30+42+48}{3} = 40$$

Sum squared error: Cluster 1 = 180, Cluster 2 = 168, Total = 348

(b) Both sets of centroids represent stable solutions, as the centroids do not change after the first iteration of the K-means algorithm.

(2) Suppose that we need to group the following eight points into three clusters:

A:(2,10), B:(2,5), C:(8,4), D:(5,8), E:(7,5), F:(6,4), G:(1,2), H:(4,9)

The distance function is Euclidean distance. Suppose initially we assign A, D and G as the prototype of the first, second and third cluster respectively. Use the K-means algorithm to find the three clusters and their respective centroids after the first iteration.

Answer:

Initial centroids: A(2,10), D(5,8), G(1,2)

- Cluster 1 (centroid = (2,10)): {A(2,10)}

- Cluster 2 (centroid = (5,8)): {C(8,4), D(5,8), E(7,5), F(6,4), H(4,9)}
- Cluster 3 (centroid = (1,2)): {B(2,5), G(1,2)}

Updated centroids: Cluster 1: (2,10), Cluster 2: $(\frac{8+5+7+6+4}{5}, \frac{4+8+5+4+9}{5}) = (6, 6)$, Cluster 3: $(\frac{2+1}{2}, \frac{5+2}{2}) = (1.5, 3.5)$

Tutorial 8 (Hierarchical Clustering)

We consider points $A(0, 0)$, $B(2, 0)$, $C(2, 4)$, $D(2, 5)$ and $E(5, 5)$.

Their distance matrix is as follows:

	A	B	C	D	E
A	0	2	$\sqrt{20}$	$\sqrt{29}$	$\sqrt{50}$
B	2	0	4	5	$\sqrt{34}$
C	$\sqrt{20}$	4	0	1	$\sqrt{10}$
D	$\sqrt{29}$	5	1	0	3
E	$\sqrt{50}$	$\sqrt{34}$	$\sqrt{10}$	3	0

Perform agglomerative hierarchical clustering on the data points using the complete link approach. Show the order in which the points are merged.

Answer: Complete link uses the maximum distance between points in different clusters.

- (1) Initially, each point is a cluster: {A}, {B}, {C}, {D}, {E}
- (2) Merge the closest clusters: {C} and {D} (distance = 1)
- (3) Merge the next closest clusters: {A} and {B} (distance = 2)
- (4) Merge the next closest clusters: {C, D} and {E} (distance = (C, E) = $\sqrt{10}$)
- (5) Merge the last two clusters: {A, B} and {C, D, E}

Exercise 2 (Association Analysis)

(1) Consider the data set shown in the following table:

Customer ID	Transaction ID	Items Bought
1	0001	{a,d,e}
1	0024	{a,b,c,e}
2	0012	{a,b,d,e}
2	0031	{a,c,d,e}
3	0015	{b,c,e}
3	0022	{b,d,e}
4	0029	{c,d}
4	0040	{a,b,c}
5	0033	{a,d,e}
5	0038	{a,b,e}

(a) Compute the support for itemsets {e}, {b,d} and {b,d,e} by treating each transaction ID as a market basket.

(b) Use the results in part (a) to compute the confidence for the association rules $\{b,d\} \rightarrow \{e\}$ and $\{e\} \rightarrow \{b,d\}$.

(c) Repeat part (a) by treating each customer ID as a market basket.

(d) Use the results in part (c) to compute the confidence for the association rules $\{b,d\} \rightarrow \{e\}$ and $\{e\} \rightarrow \{b,d\}$.

Answers:

The support for an itemset is the proportion of transactions that contain the itemset.

(a) Treating each transaction ID as a market basket:

- Support for $\{e\}$: $\sigma(e) = \frac{8}{10} = 0.8$
- Support for $\{b,d\}$: $\sigma(b, d) = \frac{2}{10} = 0.2$
- Support for $\{b,d,e\}$: $\sigma(b, d, e) = \frac{2}{10} = 0.2$

(b) Confidence for the association rules:

- Confidence for $\{b,d\} \rightarrow \{e\}$: $c(b, d \rightarrow e) = \frac{\sigma(b, d, e)}{\sigma(b, d)} = \frac{0.2}{0.2} = 1$
- Confidence for $\{e\} \rightarrow \{b,d\}$: $c(e \rightarrow b, d) = \frac{\sigma(b, d, e)}{\sigma(e)} = \frac{0.2}{0.8} = 0.25$

(c) Treating each customer ID as a market basket:

Customer ID	Items Bought
1	{a,b,c,d,e}
2	{a,b,c,d,e}
3	{b,c,d,e}
4	{a,b,c,d}
5	{a,b,d,e}

- Support for $\{e\}$: $\sigma(e) = \frac{4}{5} = 0.8$
- Support for $\{b,d\}$: $\sigma(b, d) = \frac{5}{5} = 1$
- Support for $\{b,d,e\}$: $\sigma(b, d, e) = \frac{4}{5} = 0.8$

(d) Confidence for the association rules:

- Confidence for $\{b,d\} \rightarrow \{e\}$: $c(b, d \rightarrow e) = \frac{\sigma(b, d, e)}{\sigma(b, d)} = \frac{0.8}{1} = 0.8$
- Confidence for $\{e\} \rightarrow \{b,d\}$: $c(e \rightarrow b, d) = \frac{\sigma(b, d, e)}{\sigma(e)} = \frac{0.8}{0.8} = 1$

(2) Let c_1 , c_2 and c_3 be the confidence values for the rules $\{p\} \rightarrow \{q\}$, $\{p\} \rightarrow \{q,r\}$ and $\{p,r\} \rightarrow \{q\}$ respectively. If we assume that c_1 , c_2 and c_3 have different values, what are the possible relationships that may exist among c_1 , c_2 and c_3 ?

Answer:

$$c_1 = \frac{\sigma(p, q)}{\sigma(p)}$$

$$c_2 = \frac{\sigma(p, q, r)}{\sigma(p)}$$

$$c_3 = \frac{\sigma(p, q, r)}{\sigma(p, r)}$$

Given the definition of support, the support of an itemset cannot be greater than the support of any of its subsets, i.e. $\sigma(p, q, r) \leq \sigma(p, q)$ and $\sigma(p, q, r) \leq \sigma(p, r)$. Therefore, we have:

$$c_2 \leq c_1 \text{ and } c_2 \leq c_3$$

Thus, c_2 is the smallest among the three confidence values.